Strand 4 · Data Analysis and Personal Financial Literacy

This strand (about 8 of 40 questions) has two halves: **reading data** (scatterplots, trend lines, spread) and **personal finance** (interest, saving, the cost of borrowing).

Standards covered:

- ○ **8.5C** — Contrast bivariate data that suggest a linear relationship with data that do not. ★ **8.5D** — Use a trend line to make predictions.
- ○ **8.11A** — Construct a scatterplot and describe the association. ○ **8.11B** — Mean absolute deviation (≤10 points).
- ○ **8.12A** — How rate and loan length affect the cost of credit. ○ **8.12C** — How regular saving grows. ★ **8.12D** — Simple vs. compound interest. ○ **8.12G** — Estimate college cost and a savings plan.

Part A · Scatterplots and Association (8.5C ○, 8.11A ○)

A **scatterplot** graphs paired (bivariate) data. The pattern shows the **association**: **positive linear** (points rise), **negative linear** (points fall), **non-linear** (a curve), or **no association** (scattered).

Worked example. Study time up, scores up → points rise right → **positive linear.** **Try It.** TV hours up, grades down → *negative linear.*

Part B · Trend Lines and Predictions (8.5D ★)

A **trend line** (line of best fit) runs through the middle of the data; use it to **predict** values you did not measure.

Worked example. Trend line y = 2x + 3 (x = weeks). Height at week 10: y = 2(10) + 3 = **23 cm.** **Watch Out:** a prediction is an **estimate**, not exact.

Part C · Mean Absolute Deviation (MAD) (8.11B ○)

MAD = the **average distance of each value from the mean.** Steps: (1) find the mean, (2) find each distance from the mean, (3) average those distances.

Worked example. Data 2, 4, 6, 8. Mean = 5. Distances 3, 1, 1, 3. MAD = 8 ÷ 4 = **2.** **Try It.** MAD of 10, 10, 20, 20? *(Mean 15; MAD = 5.)*

Part D · Personal Financial Literacy (8.12A/C/G ○, 8.12D ★)

Simple interest: $I = Prt$. **Compound interest:** $A = P(1 + r/n)^{nt}$ — interest earns interest, so it grows faster.

Worked example (8.12D). $500 at 4% for 3 years, **simple**: $I = 500 \times 0.04 \times 3 =$ **$60.** Compound would earn slightly more.

- **8.12A — Cost of credit.** A higher rate or a longer loan means you pay **more** interest overall.
- **8.12C — Saving regularly.** Small, steady amounts add up and grow — especially with compound interest and time.
- **8.12G — College savings.** Estimate total cost, subtract expected family contribution, divide the rest across the months you have.

Try It. Simple interest on $1,000 at 5% for 2 years? *(1000 × 0.05 × 2 = $100.)*

Strand 4 Checklist

- ☐ 8.5C, 8.11A Read scatterplots and name the association
- ☐ 8.5D Use a trend line to predict · ☐ 8.11B Calculate mean absolute deviation
- ☐ 8.12A/C/D/G Simple vs. compound interest, cost of credit, saving and college planning

Process Standards in Action (8.1 A–G)

The **mathematical process standards** are not a separate topic — they are *how* you do every problem. STAAR builds them into questions across all four strands.

Standard	What it asks you to do	How you use it
8.1A	Apply math to everyday life, society, work	Solve real-world problems (cost, distance, savings)
8.1B	Use a problem-solving model	Analyze, plan, solve, check reasonableness
8.1C	Choose the right tools and techniques	Calculator, mental math, estimation, reference sheet
8.1D	Communicate with multiple representations	Move between symbols, tables, graphs, words
8.1E	Create and use representations	Draw a diagram, make a table, sketch a graph
8.1F	Analyze relationships to connect ideas	See slope, tables, equations as the same line
8.1G	Justify reasoning with precise language	Explain *why* an answer is correct

Key Idea: Strong test-takers do more than "get the answer." They **check that it makes sense,** choose efficient tools, and can explain their thinking. Practice that on every item.

Strategy: After each answer ask: *Does this make sense?* A volume should be positive, a probability between 0 and 1, a slope matching the line's direction. Catching an unreasonable answer saves easy points.

Warm-Up Skill Drills

Short, mixed drills to warm up before the full-length tests. Do a set, then check the answer key. Each item names the TEKS it practices.

Strand 1 — Numerical Representations and Relationships

1. **(8.2D)** Order from least to greatest: 0.45, ⅖, √0.2, 43%.
2. **(8.2B)** Between which two whole numbers is √72?
3. **(8.2C)** Write 0.00056 in scientific notation.
4. **(8.2A)** Is √81 rational or irrational?
5. **(8.2C)** Write 3.2×10^5 in standard form.

Strand 2 — Computations and Algebraic Relationships

6. **(8.4C)** Find the slope of the line through (2, 3) and (6, 11).
7. **(8.5I)** A tank holds 50 L and drains 4 L per minute. Write an equation for amount y after x minutes.
8. **(8.5G)** Is {(2, 4), (3, 9), (2, 5)} a function?
9. **(8.8C)** Solve $5x - 2 = 3x + 8$.
10. **(8.5F)** Is $y = 6x$ proportional or non-proportional?

Strand 3 — Geometry and Measurement

11. **(8.7C)** A right triangle has legs 9 and 12. Find the hypotenuse.
12. **(8.7A)** Volume of a cylinder with radius 5, height 4 ($\pi \approx 3.14$).
13. **(8.10C)** Reflect (−3, 7) over the y-axis.
14. **(8.3C)** Dilate (4, −6) by scale factor ½ about the origin.
15. **(8.8D)** Two angles of a triangle are 42° and 73°. Find the third.

Strand 4 — Data Analysis and Personal Financial Literacy

16. **(8.12D)** Simple interest on $500 at 4% for 3 years.
17. **(8.11B)** Mean absolute deviation of 2, 4, 6, 8.
18. **(8.5D)** Trend line $y = 2x + 3$. Predict y when x = 10.
19. **(8.11A)** Points rise from lower-left to upper-right — name the association.
20. **(8.12A)** True or false: a longer loan at the same rate usually costs more in total interest.

Answer Key — Warm-Up Drills

1. ⅖ (0.40), 43% (0.43), √0.2 (≈0.447), 0.45 · 2. 8 and 9 · 3. 5.6×10^{-4} · 4. Rational (= 9) · 5. 320,000 · 6. m = 2 · 7. y = −4x + 50 · 8. No (input 2 → two outputs) · 9. x = 5 · 10. Proportional · 11. c = 15 · 12. 314 cubic units · 13. (3, 7) · 14. (2, −3) · 15. 65° · 16. $60 · 17. 2 · 18. 23 · 19. Positive linear · 20. True